

Your brain can start changing after just two minutes of this mental practice

New research reveals that meditation can trigger measurable brain wave changes in as little as two minutes.

Scientists used EEG tests to track 103 participants and found that these changes peaked around seven minutes of meditation.

The study showed increases in brain waves linked to relaxation and focus while decreases suggested less mind-wandering.

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You aren't alone: Many people who've sat down to meditate instead feel that their mind starts to wander within seconds. They can't relax and instantly want to give up.

But research suggests that if you can hang in there for just a few minutes, your brain could start to shift in meaningful ways – and those changes may peak after only about seven minutes of meditation.

"We see this peak last up until 15 minutes," said Dr. Balachundhar Subramaniam, professor of anesthesiology at Harvard Medical School, who co-led the study published earlier this year in the journal *Mindfulness*.

Meditation, often described as a practice of mind-body integration, involves giving your brain a break from constant noise. There are different types of meditation; the new study focused on one kind called breath-watching, a focused-attention meditation, which might mean staying in a state of calm and focusing on your breath, repeating a prayer or mantra or just sitting quietly and letting thoughts pass without chasing them.

Meditation has grown in popularity across the United States, with an estimated 1 in 5 adults – roughly 60.5 million people – reporting that they practiced meditation in 2022.

A growing body of research finds that meditation may help reduce anxiety, manage stress, ease pain and improve sleep quality, among other health benefits. The new study suggests that shifts in brain waves could play a hidden role in generating these benefits, and just a few minutes can start calming your brain.

The study included 103 adults who were asked to meditate while their brain activity was monitored using an electroencephalogram or EEG test, a tool that measures electrical patterns in the brain through sensors placed on the scalp.

The participants had a wide range of meditation experience, from beginners to advanced practitioners, and they were instructed to focus on their breathing while meditating. The EEG findings showed that significant changes in brain waves emerged within just a couple of minutes of meditation.

The devices detected increases in brain waves linked to relaxation and focus – theta, theta-alpha, alpha and beta-1 brain waves – while measuring decreases in delta and gamma-1 brain waves, which suggest less drowsiness and less mind-wandering. The data showed no change in beta-2 brain wave activity.

“You might have heard that meditation ‘boosts gamma waves,’ a type of fast brain rhythm often linked to heightened awareness. Our study found something that sounds like the opposite: one slice of gamma activity actually went down as people kept meditating,” Subramaniam said.

“That isn’t the contradiction it seems. The famous studies that found gamma going up looked at very experienced meditators, used different practices, and measured the brain in different ways than we did,” he said. “We studied something quieter: people sitting with eyes closed, simply watching their breath, with nothing to look at and no task to react to. In that calmer setting, a small dip in fast brain activity likely reflects the mind settling – less wandering, fewer distractions – as attention steadies over the first several minutes.”

What moved in the brain

- ↑ Alpha (associated with calm wakefulness)
- ↑ Beta-1 (associated with focused engagement)
- ↓ Gamma-1 (associated with less mind-wandering)
- ↓ Delta (associated with less drowsiness)
- ↑ Theta (associated with internalized attention)
- ↑ Theta-alpha (associated with meditative coactivation)

As a whole, these changes in the study participants’ brains suggest that their minds were moving into calmer, more attentive states in mere minutes.

“When there is co-activation of theta, alpha and theta-alpha, there is more attention to yourself. So it is almost like a flashlight within you,” Subramaniam said. Along with internalized attention, increases in these brain waves are associated with feelings of calm and creativity.

Activations of beta-1 waves can mean “you are more aware and more awake,” he said, describing this state as “relaxed alertness,” in which someone may feel more focused. Decreases in delta and gamma-1 brain waves also can be associated with “relaxed alertness.”

The researchers also found that, compared with novice meditators, people who were more advanced showed even greater increases in theta and theta-alpha brain waves and larger drops in the delta and gamma-1 brain waves.

That finding suggests that like any skill, the potential effects of meditation can deepen with practice.

Subramaniam said he often hears two main pushbacks from people who are hesitant to try meditation: They don’t have the time, or they can’t stop their mind from wandering.

But the new study suggests that “at the seventh minute or so, if you sit quietly, then your breath-watching will happen automatically, the way it is intended to be. That’s a very simple way of getting into the space that everyone is aspiring to be,” said Subramaniam, who was also involved in the recent launch of the free Miracle of Mind guided-meditation app, developed by an international team of volunteers at the Isha Foundation.

“If you’re a novice and you’re starting for the first time, doing a guided meditation is the best way to start. You just follow instructions,” he said. “When you do that over a period of time, let’s say four to six weeks or so, then now it is becoming your habit.”

Ultimately, the new study not only reveals that meditation can be associated with changes in brain waves, it highlights when those changes begin and how they build over time, said Dr. Ignacio Saez, director of the Laboratory for Human Neurophysiology at the Icahn School of Medicine at Mount Sinai in New York.

“I think meditation changes brainwaves because it changes what the brain is doing,” Saez, who was not involved in the new study, said in an email. “When someone moves from ordinary mind-wandering into a deliberate practice like breath awareness or loving-kindness, the brain is reallocating resources toward attention, self-monitoring, emotion regulation, and sometimes memory-related processes. Those are changes in the brain’s state, and should leave an electrical signature.”

In a study published last year, Saez and his colleagues used surgical EEG recordings to identify that meditation can lead to changes in brain waves in the amygdala and hippocampus, key brain regions involved in emotional regulation and memory.

“So, to me, the common message is not that meditation produces one fixed brainwave pattern, but that it reorganizes activity across a wider brain network,” he wrote. “There probably is not one universal ‘meditation wave.’”

The new research sheds light on what happens in the brain within a single meditation session — and it comes as no surprise that changes appear within minutes, said Alexander Duda, an associate research fellow at the University of Wollongong who has studied how mindfulness meditation may impact activity in the brain but who was not involved in the new study.

“It makes sense that neurophysiological, and other physiological, changes could appear within just a few minutes,” Duda said in an email. But he added that more research is needed to help explain what these changes may mean for health outcomes.

There is evidence linking meditation-related changes with cardiovascular measures, for instance, “but the current evidence remains limited and developing, and any single neurophysiological change is unlikely to fully explain a specific health outcome on its own,” he said.

Saez also said that he would be careful not to treat a shift in brain waves as being a health outcome by itself.

“These EEG changes are better understood as markers that the meditator has entered a different functional state,” Saez said.

“In the new study, the pattern is interpreted as a kind of relaxed alertness, and the deeper-brain findings in my work suggest that regions involved in emotion and memory are participating in that shift as well,” he said. “That is meaningful, but it is not the same thing as proving improved blood pressure, mood, or brain structure from one short session alone.”

He added that these EEG changes in the brain may serve as “a window into how meditation may be working by affecting brain activity,” rather than as a stand-alone health biomarker.